

CARDANO-KERI · IDENTITY CORE

Keep the identity you already trust.

The first cardano-keri milestone delivers the identity core: an existing KERI identity gains a stable home on Cardano, so a team can manage records, update keys, and recover control without handing authority to a middleman.

• WIRE CONTRACT READY AND BYTE-TESTED

• FIRST MILESTONE ADDS VALIDATOR + SAFEGUARDS

Meet Northstar: a team with an identity already in use.

Northstar wants its existing KERI identity to control Cardano records, so partners can recognize the same organization on both systems.

WHAT THEY ALREADY USE

A standard KERI identity

AID: E... • Blake3 • Veridian-compatible

Three officers share control: any two can approve an action. A separate set of future keys is kept safe for rotation.

WHAT THEY WANT ON CARDANO

One record that follows the organization

Partners should always find Northstar’s current authorized keys through one stable identity reference.

identity → authority → data

WHAT THEY SHOULD NOT HAVE TO DO

Start over or trust a middleman

No replacement identity, no permanently fixed hot key, and no bridge custodian in the everyday authorization path.

Bring your existing identity—no conversion required.

Northstar registers the same **E-prefix KERI identifier** its partners already know. There is no Cardano-only version of the identity and no wallet-vendor lock-in.



WHAT IS VERIFIED Cardano proves that the inception-event bytes produce this AID, and the signed registration proves control of the event's keys. Translating KERI fields into the checkpoint is separately attested and can be challenged.

same identityNo re-issuance and no Cardano-specific AID.

stable handleOne quantity-one token is derived from the AID.

on-chain proofEvents up to 1024 bytes are bound to the AID in script.

Give apps one stable name for your organization.

A Cardano app can connect to Northstar once. The identity reference stays the same while the authorized keys and approval rules change over time.

NEVER CHANGES

Stable identity token

policy_id

+

aid_asset_name

quantity = 1 • stays the same across rotations

WHAT THE CHECKPOINT KEEPS CURRENT

current keys	The Ed25519 keys allowed to act today, with their approval rule.	k / kt
next keys	The public commitment that makes the next safe rotation possible.	n / nt
witnesses	The witnesses and receipt threshold that confirm the public history.	b / bt
position	Where this checkpoint sits in both Cardano and the KERI event log.	seq / s

one lookup Apps resolve the checkpoint by policy and asset.

one format The versioned datum is frozen and byte-tested.

continuous The handle is identical before and after rotation.

Keep the approval rules your team already uses.

Northstar does not need to simplify its governance for Cardano. The checkpoint carries the same signing rule the team uses in its KERI identity.

NORTHSTAR'S EVERYDAY RULE

key 1
finance lead

key 2
operations lead

key 3
director

Any two must sign

kt = 2

WEIGHTED APPROVALS

Give people halves, thirds, or other fractional voting weights.

MORE THAN ONE GROUP

Require separate clauses and keep zero-weight reserve positions.

SOLO BUILDERS INCLUDED

1-of-1 uses the same checkpoint format.

- KERI-faithful

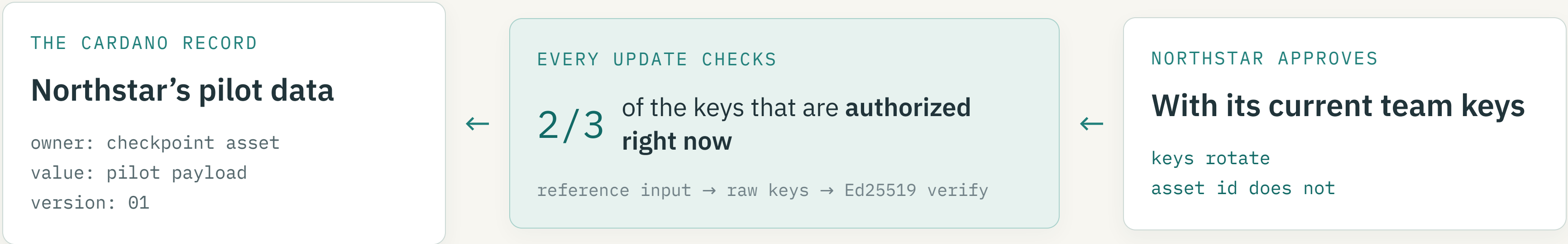
The rules match the reference implementation.
- production-tested

Observed GLEIF and QVI configurations round-trip.
- efficient

Everyday authorization uses raw keys and no hashing.

Your Cardano records stay under the right control—even after keys change.

Northstar owns an on-chain record through its identity checkpoint. The record follows the organization’s current authority instead of being tied to one permanent key.



After a rotation, the record and owner stay the same; only the authorized keys change.

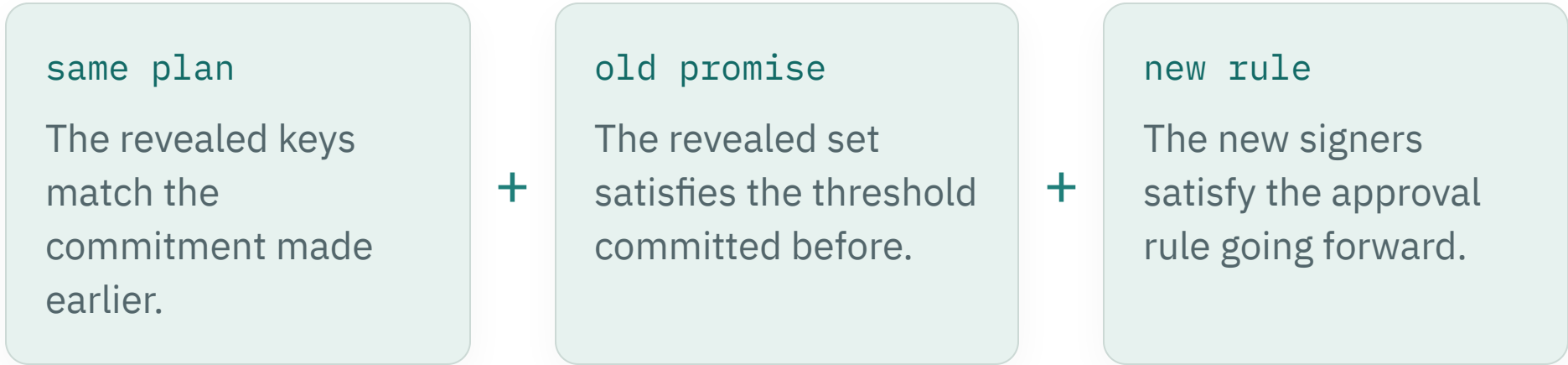
NO KEY MIGRATION

current authority Every write checks the live checkpoint. stable ownership The record does not move when keys rotate. end to end The runnable demo writes a real Cardano record through the identity.

Update keys directly, whenever your team needs to.

Northstar can replace operational keys itself. A valid rotation follows the plan committed in advance, without waiting for an administrator, allowlist, or oracle.

THREE CHECKS PROTECT EVERY KEY UPDATE



The checkpoint moves forward to `seq + 1` SELF-SERVICE

PRODUCTION RESERVE EXAMPLE

Use 3 of 7 prepared keys

GLEIF Root uses this pattern: reveal only the keys needed now and keep the remaining reserve keys private.

3.6%

CPU / reveal

4.5%

mem / reveal

- direct

The identity controller submits its own valid update.
- flexible reserves

A rotation can reveal only part of the prepared set.
- measured cost

Each revealed key uses one small Blake3 block.

Losing today’s keys does not mean losing the identity.

Even if every key valid today is stolen, the intruder cannot choose tomorrow’s keys. Northstar can use the successor keys it prepared earlier to recover control.

WHAT THE INTRUDER HAS

Every key that is valid today

today’s signing rule is satisfied

✗ Cannot choose tomorrow’s keys



promise about
the next keys

WHAT NORTHSTAR KEPT SAFE

The keys prepared for the next rotation

$\text{blake3}(\text{qb64}(\text{key})) \in \text{prior } n$

✓ Recovers the same identity

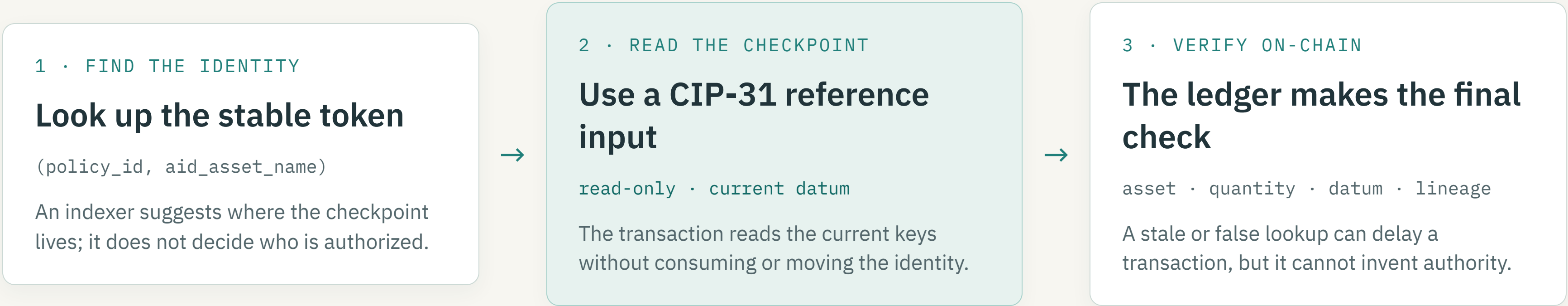
tested failure A full stolen current quorum still cannot rotate.

continuity Successor keys remove the compromised set without changing the AID.

scope boundary If both current and successor keys are lost, this layer cannot recover the identity.

Apps always read the keys that are valid now.

A Cardano app uses one read-only reference to Northstar’s checkpoint. The ledger verifies the identity record again before it accepts an action.



Safe by default: if the identity is paused, applications reject actions automatically until the checkpoint catches up.

FAIL CLOSED

simple integration The app performs one lookup and one reference read.

limited trust An indexer can delay service, but it cannot forge authority.

rotation-safe The same asset resolves the new keys after an update.

Public checks keep Cardano and KERI in sync.

Partners do not have to take Northstar’s word for it. Public evidence tells the validator when to retire a conflicting identity or pause an out-of-date checkpoint.

TWO CONFLICTING ROTATIONS

The owner signed two different futures

Retire the identity

The validator burns the token, leaves a permanent tombstone, and pays the prover from the deposit.

KERI HAS MOVED FIRST

A witnessed KERI rotation is newer

Pause the checkpoint

The identity can only move forward until Cardano catches up. Applications reject authority in the meantime.

CARDANO APPEARS AHEAD

Missing public history cannot be proved

Require receipts

Witness receipts are required before an advance. Later KERI events resolve the state toward agreement or conviction.

Owners cannot be framed: witness signatures alone are not enough. Conviction requires the owner’s pre-committed signing threshold.

PART OF THE IDENTITY-CORE VALIDATOR

The same public evidence supports useful services.

Because the KERI history and Cardano checkpoint are public, independent operators can keep them aligned, audit them, and publish credential status.

WATCHER

Keep both histories aligned.

Compare the public KERI event log (KEL) with Cardano. Submit valid pause or conviction evidence and receive the bounty.

open to anyone
mechanical comparison
paid from the deposit

AUDITOR

Build reports from replayable facts.

Reconstruct the full KERI history, compare it byte-for-byte with Cardano, and cite the evidence.

full history replay
byte-for-byte evidence
retirement remains visible

ISSUER

Publish credential status on-chain.

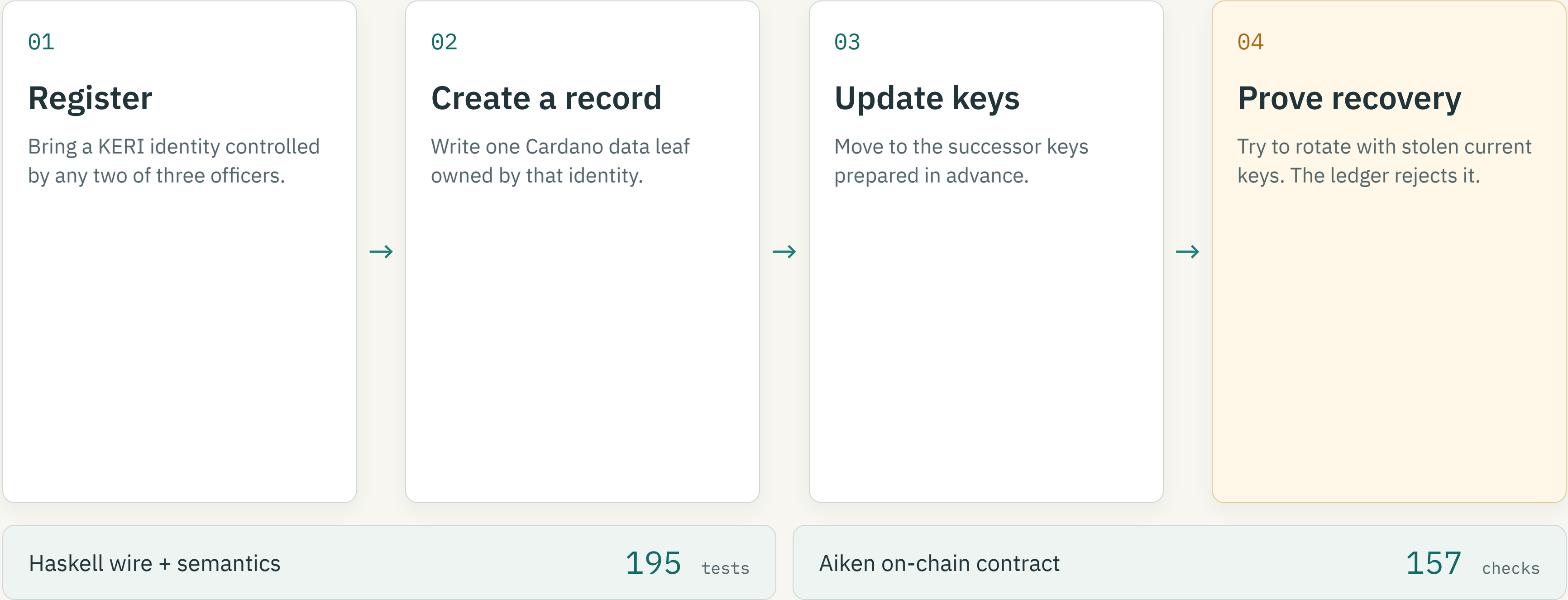
Mirror each issuer’s revocation registry so consumers can check status without calling the issuer’s API.

production credential IDs
ledger-ordered updates
issuer API not required

CLEAR SCOPE The identity core publishes credential status. Verifying the full GLEIF → QVI → legal-entity credential chain comes in the next verification milestone.

The end-to-end demo shows the complete journey.

A runnable local-devnet demo proves that registration, record ownership, key updates, and recovery work together—not only as separate unit tests.



What this foundation proves—and what comes next.

DELIVERED BY THE IDENTITY CORE

- ✓ Use an existing E-native KERI identity without re-issuing it.
- ✓ Keep one stable Cardano handle while keys and approval rules change.
- ✓ Own records, replay evidence, manage lifecycle, and publish credential status.
- ✓ Pause or retire a checkpoint when public evidence proves divergence.

BUILT ON TOP IN LATER MILESTONES

- Verify the complete GLEIF → QVI → legal-entity credential chain.
- Authorize arbitrary dApp actions through reusable signed envelopes.
- Connect Veridian and Signify signing directly to Cardano workflows.
- Add delegated identities and a superseding recovery path.

A USEFUL FIRST PILOT NEEDS

an existing KERI identity · one Cardano record it should control · a clear freshness policy

[read the engineering deep dive ↗](#)

[bring us a real workflow →](#)